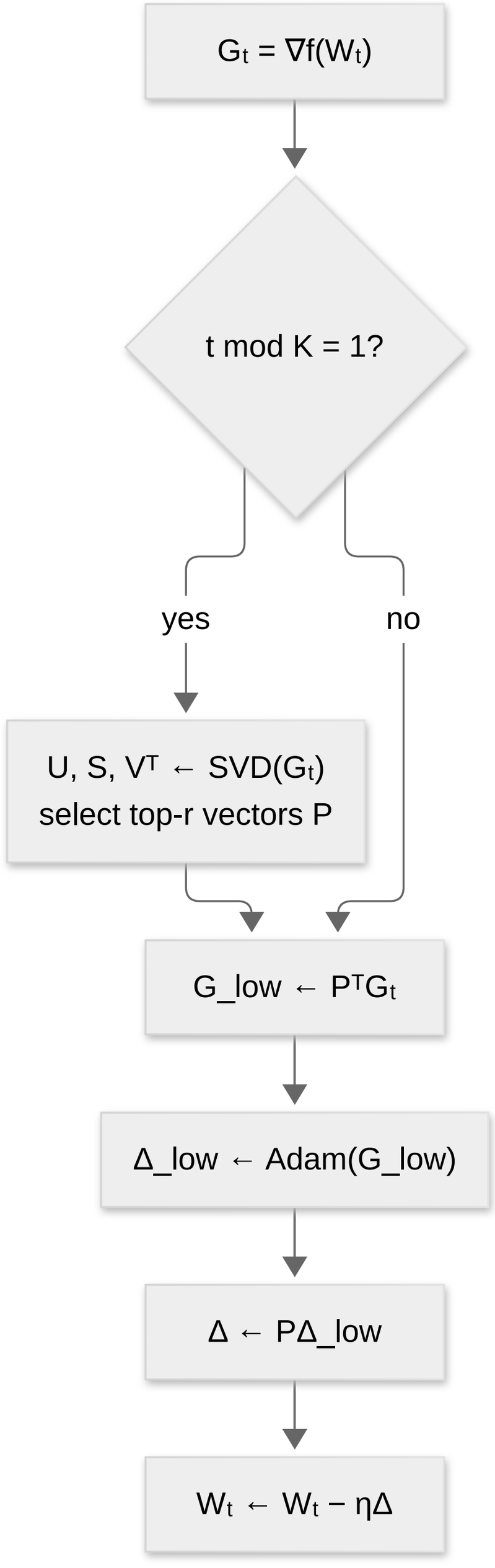

$G_t = \nabla f(W_t)$
t mod K = 1?
yes
no
U, S, $V^T \leftarrow$ SVD(G_t)
select top-r vectors P
$G_low \leftarrow P^T G_t$
$\Delta_low \leftarrow Adam(G_low)$
$\Delta \leftarrow P\Delta_low$
$W_t \leftarrow W_t - \eta\Delta$